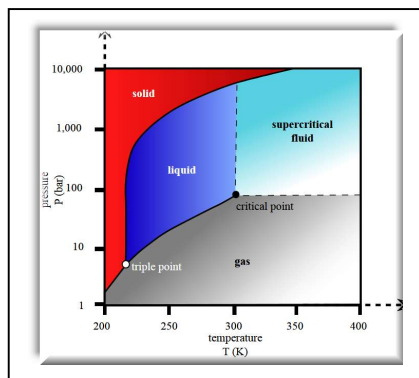




College:
Department:
Course:
Classroom and Time:

UIC COLLEGE OF ENGINEERING
UIC Department of Chemical Engineering
CHE 301: Chemical Engineering Thermodynamics
EIB 271 M|W: 11:00 AM – 12:15 PM

Instructor:
Office hours:
E-mail:
Cell Phone:

Dr. Michael Caracotsios, EIB 244
M|Tu|W|Th: Stop by my office or schedule a ZOOM meeting
mcaracot@uic.edu
(630) 696-0934

TA Name:
E-mail:
Office hours:

Samagra Dvivedi, EIB 264
sdviv@uic.edu
M: 2:00PM- 4:00PM |Th: 10:00AM – 12:00 Noon

COURSE OBJECTIVES

This course is designed to extend the student's acquired knowledge and understanding of the first and second law of thermodynamics to the most critical areas of Chemical Engineering Thermodynamics and Chemical Reaction Equilibrium. These areas encompass properties of pure fluids and fluids in homogeneous solutions, single and multicomponent phase equilibrium, and chemical equilibrium. The knowledge of this material will provide a strong foundation for all subsequent courses in chemical engineering and contribute towards understanding of chemical and physical processes. By the end of the course the students should be able to:

- ✚ Understand the basic concepts of the five thermodynamic potentials: Internal Energy, Gibbs and Helmholtz Free Energy, Entropy and Enthalpy.
- ✚ Apply control volume analysis to a variety of situations related the phase and chemical equilibrium and solve problems utilizing computer software.
- ✚ Understand the theory of thermodynamics of ideal and non-ideal solutions
- ✚ Understand the applicability and use of various Equations of State in describing the thermodynamic state of pure fluids and mixtures thereof.
- ✚ Understand the applicability and use of various Liquid Activity Coefficient Methods in describing the thermodynamic state of non-ideal liquid mixtures
- ✚ Determine equilibrium conditions between phases using equations of state and liquid activity coefficients.
- ✚ Determine equilibrium conditions between reactants and products using equilibrium constants and Gibbs energy minimization concepts.

COURSE PREREQUISITES

The following are key requirements for a successful completion of this course.

- **Mathematics:** Differential and Integral Calculus
- **Statistics:** Linear and Nonlinear Regression
- **Numerical Methods for Chemical Engineering**
- **Programming Experience:** Excel VBA, Matlab, C++
- **CHE Department Prerequisites:** CHE 201, CHE 205 and CHE 210

REQUIRED SOFTWARE

All students must download and install the Microsoft EXCEL [32-bit Version] **CHE Calculator v12.0** following the instructions listed on the UIC Blackboard in the **Computational Tools** content area of the course.

REQUIRED TEXTBOOK

Michael Caracotsios, *An Introduction to Chemical Engineering Thermodynamics. Theory and Applications*, 1st Edition, 2025 will be distributed to all students free of charge. Additional material in the **Course Textbook** content area of the course.

RECOMMENDED TEXTBOOKS

1. Themis Matsoukas, *Fundamentals of Chemical Engineering Thermodynamics*, Prentice Hall, 2013.
2. J. M. Smith, H. C. Van Ness and M. M. Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th Edition, McGraw-Hill, 2005.
3. S. I. Sandler, *Chemical and Engineering Thermodynamics*, 3rd Edition, John Wiley & Sons, Inc., 1999.

CHE 301: COURSE OUTLINE**1. CHAPTER 1: REVIEW**

- Mass and Energy Balances.
- Momentum and Mechanical Energy Balances.
- Entropy and Lost Work Balances.
- Calculus and Numerical Methods for Chemical Engineers

2. CHAPTER 2: APPLICATIONS OF MACROSCOPIC BALANCES

- Thermodynamic Cycles.
- Steam Purging from Pressure Tanks.
- Other Applications from Chemical Engineering.

3. CHAPTER 3: THERMAL PROPERTIES OF PURE FLUIDS

- Thermodynamic Potentials
- Review: Maxwell Equations
- Thermal Properties of Pure Fluids: Ideal Gases and Liquids
- Vapor-Liquid Equilibrium between Ideal Gases and Liquids

4. CHAPTER 4: EQUATIONS OF STATE FOR PURE FLUIDS

- Cubic and Virial Equations of State
- Residual Properties for Enthalpy, Entropy and Heat Capacity
- Fugacity and Fugacity Coefficients from Equations of State
- Thermal Properties of Pure Fluids with Equations of State
- Vapor-Liquid Equilibrium Fundamentals using Equations of State

5. CHAPTER 5: THERMAL PROPERTIES OF MIXTURES

- Partial Molar Properties
- Thermal Properties of Ideal Gas and Liquid Mixtures
- Gibbs Thermodynamic Consistency Theorem
- Vapor Liquid Equilibrium for Ideal Gas and Liquid Mixtures
- Bubble point, Dew Point and Pressure-Temperature Flash Calculations

6. CHAPTER 6: EQUATIONS OF STATE FOR MIXTURES

- Cubic Equations of State and Mixing Rules
- Thermal Properties of Non-ideal Gas and Liquid Mixtures
- Fugacity and Fugacity Coefficients of Components in Mixtures
- VLE Calculations for Non-Ideal Gas and Liquid Mixtures
- Bubble point, Dew Point and Pressure-Temperature Flash Calculations

7. CHAPTER 7: LIQUID ACTIVITY COEFFICIENTS

- Gibbs Generating Excess Functions
- Liquid Activity Coefficients (NRTL, van Laar, Wilson, Margules)
- Bubble Point, Dew Point and Pressure-Temperature Flash Calculations
- Azeotrope Finding Calculations
- VLE Data Reduction Methods and Thermodynamic Consistency Tests

8. CHAPTER 8: CHEMICAL REACTION EQUILIBRIUM

- Criteria for Equilibrium in Chemical Reactions
- Gibbs Free Energy and Equilibrium Constants
- Evaluation of Equilibrium Constants from Thermodynamic Data
- Calculation of Equilibrium Composition for Single and Multi-Reaction Systems

CHE 301: COURSE GRADING

Your final grade for the **CHE 301** Chemical Engineering Thermodynamics course will be determined from (**max 10**) take home assignments, two midterms and one final examination, and (**max 10**) quiz/attendance records. The contribution of all these activities to the grade calculation formula is shown below:

1. Home Assignments: [50%]
2. Examinations: [30%]
3. Quizzes & Attendance: [20%]

The formula for the calculation of your total score is given by the equation below. The grades associated to a total score range are also given in the table below.

$$TS(\text{Total Score}) = 0.50\langle H \rangle + 0.30\langle E \rangle + 0.20\langle Q + A \rangle$$

Total Score	Grade	Symbol	Max Score	Meaning of Symbols
$90 \leq TS \leq 100$	A			
$80 \leq TS < 90$	B	$\langle H \rangle$	100 pts	Average Score for Home Assignments
$70 \leq TS < 80$	C	$\langle E \rangle$	100 pts	Average Score for Examinations
$60 \leq TS < 70$	D	$\langle Q + A \rangle$	100 pts	Average Score for Quizzes & Attendance

Note 1: Completion of all Assignments is mandatory. In the event that a student does not complete an Assignment by the designated time and date, the student will receive a score of -100 points. The student may still be able to negotiate a better score by turning in the missed assignment at a time agreed upon with me. If attendance is taken and a student is absent the student will receive a score of -100 points, unless there is a reasonable explanation for the absence.

Note 2: Extra credit may be given at my discretion. The extra credit will be based on the student's record of attendance, punctuality, participation in class, good work and ethics behavior, and reasonable scores on all assignments and examinations. The extra credit will be contingent on perfect attendance and completion of all home assignments.

Dr. Michael Caracotsios
 CLINICAL ASSOCIATE PROFESSOR
 CHEMICAL ENGINEERING DEPARTMENT
 UNIVERSITY OF ILLINOIS CHICAGO

PHONE: +1 (630) 696-0934
 Email: mcaracot@uic.edu

CHE 301: MISCELLANEOUS

Miscellaneous information about this course, the policy on partial credit as well as the policies on student courtesy, tardiness and absenteeism is given in the sections below. Contact Professor Michael Caracotsios if you have any questions.

BLACKBOARD COLLABORATE HELP

By email: LTS@uic.edu

By phone: (312) 413-0003 Option #1

ACADEMIC INTEGRITY

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy: <https://dos.uic.edu/conductforstudents.shtml>

RELIGIOUS HOLIDAYS

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure. The University Holidays and the Religious Observances calendar can be found online at the following website: <http://oae.uic.edu/religious-calendar/>. The list includes national holidays recognized by the University as well as religious days of special observance that may prohibit a student from attending classes. Please keep in mind this list is not exhaustive.

PARTIAL CREDIT POLICY

Partial credit can be requested under certain circumstances. The student should make an appointment with the Professor and be prepared to discuss the reasons for requesting the extra credit. Final decision rests with the Professor.

No partial credit will be given for:

1. Incorrect formulas, full or partial.
2. Inconsistent or missing units in calculations.
3. Multiple answers.
4. Incoherent (illogical, disjointed) derivations or calculations.

STUDENT COURTESY POLICY

Talking/Smart Phones/Laptops/Food and Drinks in Class/Tardiness:

Not allowed unless explicitly stated

Repeat Offenders will be asked to leave the classroom

Once instruction begins the class doors will close

Cheating/Copying:

Perpetrators will receive zero points in their homework

Homework contribution to the final grade is substantially reduced after first warning

Late Home Assignments and Disputes:

Late Home Assignments will not be accepted without prior arrangements with the Professor.

If there are Home Assignments score disputes, they must be resolved first with the TA.

If there are Exam score disputes, they will be resolved only with the Professor.